

CHE110: Environmental Studies

100-Percentile Master Study Guide, Textbooks Deep-Dive & Curated Video Hub

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1. Standard Reference Textbooks & Deep-Dive Chapter Context

Unit	Subject Area	Standard Textbook	Core Chapter Concept & Key Exam Takeaways
Unit I & II	Ecology & Sustainability	Anubha Kaushik & C. P. Kaushik (New Age, 7th Ed)	Ch 1 & 2: 4 Spheres (Atmosphere, Hydrosphere, Lithosphere, Biosphere), Multidisciplinary scope, Brundtland 1987, 3 Pillars of Sustainability, Carrying capacity, 17 SDGs. Ch 3: Resource degradation (groundwater, dams, deforestation). Ch 4: Biotic/Abiotic structure, 10% Lindeman's Rule, Food chains/webs, Ecological Pyramids (Energy always upright, inverted aquatic biomass), Succession (Hydrosere vs Xerosere).
Unit III	Biodiversity & Conservation	Erach Bharucha (UGC / Orient BlackSwan)	Ch 4: 3 Levels (Genetic, Species, Ecosystem), 4 Ecosystem Services, Norman Myers Hotspot criteria (≥1500 endemic plants, ≥70% habitat loss), 4 Indian Hotspots (Himalaya, Western Ghats, Indo-Burma, Sundaland), HIPPO Threats, IUCN Red List, In-situ (National Parks, Sanctuaries, Biosphere Reserves, Sacred Groves) vs Ex-situ (Botanical Gardens, Seed Banks, Cryopreservation).
Unit IV	Pollution & Waste Management	D. Dave & S. S. Katewa (Cengage Learning)	Ch 5: Air Pollution (Primary vs Secondary, Classical vs Photochemical Smog, ESP/Scrubbers), Water Pollution (BOD vs COD math, Eutrophication sequence, Minamata/Itai-Itai), Ozone Chapman cycle & Montreal Protocol 1987, Acid Rain (H2SO4/HNO3), 4Rs Solid Waste hierarchy, Bhopal Gas Tragedy (MIC).
Unit V & VI	Disasters, Laws & Movements	Kaushik & Kaushik / Erach Bharucha	Disaster Mgmt: Pre-during-post disaster cycle, 3-tier framework: NDMA (PM), SDMA (CM), DDMA (DM), NDRF. Laws: Wildlife 1972, Water 1974, Forest 1980, Air 1981, EPA 1986 (Umbrella law post-Bhopal), NGT 2010. Movements: Bishnoi (1730), Chipko (1973 - Bahuguna/Gaura Devi), Appiko (1983), Silent Valley (1978), Jungle Bachao (1982), Narmada Bachao (1985 - Medha Patkar).

2. Curated YouTube Video Playlists & Online Resource Links

Unit / Topic	Recommended Channel	Search Query / Video Reference	Core Takeaway & Key Concepts
Unit I: Sustainability & 17 SDGs	CrashCourse	CrashCourse Sustainable Development Goals	Brundtland definition, 17 SDGs, Carrying Capacity.
Unit II: Ecosystems & Energy Flow	Amoeba Sisters	Amoeba Sisters Ecological Succession and Energy Flow	10% Lindeman's Rule, Food Chains vs Webs, Succession.
Unit II: Ecological Pyramids	Gate Smashers / NEET	Ecological Pyramids Number Biomass Energy	Why Energy Pyramids are always upright; inverted biomass.
Unit III: Biodiversity Hotspots	StudyIQ IAS	StudyIQ Biodiversity Hotspots in India	4 Indian Hotspots (Western Ghats, Himalayas, etc.).
Unit III: In-situ vs Ex-situ	Amit Sengupta	Amit Sengupta In situ and Ex situ Conservation	National Parks vs Sanctuaries vs Biosphere Reserves vs Seed Banks.
Unit IV: Smog, BOD & Eutrophication	StudyIQ IAS	Photochemical Smog vs Classical Smog StudyIQ	Classical vs Photochemical smog, Eutrophication sequence.
Unit V: Disaster Management (NDMA)	StudyIQ IAS	National Disaster Management Authority NDMA India	NDMA (PM), SDMA, DDMA structure, Disaster Lifecycle.
Unit VI: Environmental Movements	StudyIQ / Drishti IAS	Environmental Movements in India Chipko Narmada	Bishnoi (1730), Chipko (1973), Appiko, Silent Valley, NBA.
Unit VI: Major Environmental Acts	StudyIQ IAS	Important Environmental Acts in India EPA 1986	Wildlife 1972, Water 1974, Forest 1980, Air 1981, EPA 1986, NGT.

3. High-Yield Unit Summaries, Diagrams & Formulas

Unit I & II (Ecology & Resources): 4 Spheres (Atmosphere, Hydrosphere, Lithosphere, Biosphere). 3 Pillars of Sustainability (Environmental, Social, Economic). Carrying capacity is the max sustainable population size. Ecosystem energy flow follows Lindeman's 10% rule (90% lost as heat at each trophic step). Energy Pyramids are ALWAYS upright due to 2nd Law of Thermodynamics; Aquatic biomass pyramids are inverted. Succession stages: Pioneer → Seral stages → Climax Forest.

Unit III (Biodiversity): 3 Levels (Genetic, Species, Ecosystem). Norman Myers Hotspot criteria: ≥1500 endemic vascular plants and ≥70% habitat loss. India has 4 Hotspots: Himalaya, Western Ghats, Indo-Burma, Sundaland. In-situ (National Parks, Wildlife Sanctuaries, Biosphere Reserves, Sacred Groves) vs Ex-situ (Botanical Gardens, Zoos, Seed Banks, Cryopreservation -196°C). HIPPO threats framework.

Unit IV (Pollution & Waste): Classical Smog (reducing, cool, SO2) vs Photochemical Smog (oxidizing, sunny, NOx + VOCs + UV → Ozone + PAN). Eutrophication: Fertilizer runoff → Algal Bloom → Sunlight blocked → Plant death → Bacterial DO consumption → Dissolved Oxygen crash & Fish kill. Ozone depletion: 1 Chlorine radical from CFCs destroys >100,000 O3 molecules (Montreal Protocol 1987). 4Rs: Reduce, Reuse, Recycle, Recover.

Unit V (Disasters): Disaster cycle: Pre-disaster (Mitigation, Preparedness, Early Warning) → During (Response, Search & Rescue) → Post-disaster (Relief, Rehabilitation, Reconstruction). Indian 3-tier structure: NDMA (Prime Minister) → SDMA (Chief Minister) → DDMA (District Collector/Magistrate). Executive rescue force is NDRF.

Unit VI (Laws & Movements): Key Acts: Wildlife 1972, Water 1974 (CPCB/SPCB), Forest 1980, Air 1981 (Noise included 1987), EPA 1986 (Umbrella law post-Bhopal), NGT 2010. Historic Movements: Bishnoi 1730 (Amrita Devi), Chipko 1973 (Sunderlal Bahuguna, Gaura Devi), Silent Valley 1978 (KSSP), Jungle Bachao 1982, Appiko 1983 (Pandurang Hegde), Narmada Bachao 1985 (Medha Patkar, Baba Amte).

4. 100-Percentile Exam Presentation Blueprint

Tier 1: Formal Definition (1-2 crisp lines with technical keywords bolded)
Tier 2: Labeled Architectural Block Diagram / Flowchart (Drawn neatly in a box)
Tier 3: Structured Comparative Table / Bullet Points (Highlighting contrast & parameters)
Tier 4: Statutory Act / Case Study / Chemical Formula (Guarantees full marks from examiner)